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Certitrust

Blockchain-Anchored Digital Certificate Issuance & Cryptographic Verification — Trust the Blockchain, Not the Website.

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SHA-256	< 2s	100%	Polygon	0	DigiLocker
HASH ALGORITHM	VERIFICATION TIME	TAMPER EVIDENCE	BLOCKCHAIN NETWORK	CENTRAL SERVER DEPENDENCY	INTEGRATION READY

1 ABSTRACT

"A certificate is only as trustworthy as the system verifying it — and today's systems are built on URLs, not cryptography."

Certitrust is a full-stack digital credentialing platform that issues and verifies academic and professional certificates using **cryptographic hashing (SHA-256)** and **blockchain anchoring on Polygon Amoy**. It eliminates the single point of failure inherent in centralized verification databases by recording every certificate hash immutably on-chain. Anyone — employer, institution, or individual — can independently verify a certificate's authenticity by querying the public blockchain directly, with no dependency on Certitrust's servers.

The platform additionally supports **forensic OCR-based tamper detection** via Azure Document Intelligence, **bulk issuance via CSV**, institutional branding, and a **DigiLocker-compatible PULL API** ready for integration with India's national credential infrastructure.

2 PROBLEM STATEMENT

Current certificate verification systems suffer from three critical flaws that make credentials susceptible to forgery and system failure:

△ Spoofable QR Codes
QR codes link to centralized URLs. Cloning a website and replacing the link creates a convincing fake that "verifies" against a fraudulent database — with zero technical barrier.

△ Single Point of Failure
Centralized databases mean verification fails when the server is down, company shuts down, or DNS lapses. Trust is tied to institutional uptime, not truth.

△ No Cryptographic Proof
Visual watermarks and institutional stamps can be replicated. Without digital signatures and hashing, authenticity is an assumption — not a mathematical guarantee.

3 PROPOSED SYSTEM

Certitrust replaces the trust-the-website model with a **three-layer cryptographic trust stack**:

Layer 1 — Cryptographic Hash
Every certificate generates a unique SHA-256 fingerprint from metadata (issuer, recipient, course, date, UUID). Any modification — even one character — produces a completely different hash, making silent tampering cryptographically impossible.

Layer 2 — On-Chain Registry
The hash is recorded in **CertChainRegistry.sol** on Polygon Amoy. The issuer's wallet address and block timestamp are permanently linked. No one can insert a record without the issuer's private key.

Layer 3 — Independent Verification
Anyone can verify by querying the public smart contract — or Polygonscan directly. Even if Certitrust's website is offline or cloned, the blockchain is the ground truth.

4 METHODOLOGY

Issuance Flow

1. **Form Submission:** Issuer enters recipient name, email, course, date, certificate type — validated via Zod schema server-side.
2. **Hash Computation:** Server computes `SHA-256("issuer_id | name | course | date | uuid")` — deterministic, collision-resistant.
3. **Blockchain Anchoring:** Ethers.js calls `CertChainRegistry.issue(bytes32 hash)` on Polygon Amoy. Tx signed with deployer private key server-side.
4. **PDF Generation:** @react-pdf/renderer builds an institutional certificate (A4 landscape) embedding QR code, hash footer, org logo & signature.
5. **Storage:** PDF stored in Supabase Storage; metadata saved in PostgreSQL with tx hash, pdf URL, and DigiLocker URI.

Verification Flow

1. **Input:** Verifier scans QR code or pastes hash at `/verify` — no account required.
2. **On-Chain Lookup:** `CertChainRegistry.verify(hash)` called via read-only Ethers.js provider — zero gas cost.
3. **Result Display:** Returns issuer address, issuance timestamp, and revocation status. Green / Red / Not Found badge shown.
4. **Forensic Check (optional):** Upload certificate image/PDF → Azure Document Intelligence OCR → re-hash extracted fields → compare vs on-chain hash.

System Architecture

Presentation Layer
Next.js 16 App Router · shadcn/ui · Tailwind CSS 4 · React 19

▼

Application Layer
Next.js Server Actions · Zod Validation · @react-pdf/renderer · Azure OCR

▼

Data Layer
Supabase PostgreSQL (RLS) · Supabase Storage · Supabase Auth

▼

Blockchain Layer
CertChainRegistry.sol · Polygon Amoy (Chain ID 80002) · Ethers.js 6 · Hardhat

5 SMART CONTRACT

The core logic lives in **CertChainRegistry.sol** (Solidity ^0.8.24), deployed on Polygon Amoy. It stores a mapping of `bytes32 hash → CertRecord` with three functions:

```
struct CertRecord {
  address issuer;   // who issued
  uint256 issuedAt;  // block timestamp
  uint256 revokedAt; // 0 = active
}
mapping(bytes32 => CertRecord) public records;


issue(bytes32 hash) → stores on-chain, prevents duplicates
revoke(bytes32 hash) → only issuer, immutable timestamp
verify(bytes32 hash) → view, zero gas cost
```

6 KEY FEATURES

Cryptographic Issuance
SHA-256 fingerprint from metadata — deterministic, collision-resistant identity per certificate.

Blockchain Anchoring
Immutable Polygon Amoy on-chain registry links hash to issuer wallet and block timestamp.

Public Verification
No login required. Hash lookup via read-only smart contract call. Zero gas cost for verifiers.

Forensic OCR Check
Upload certificate image → Azure Document Intelligence extracts fields → re-hash → compare on-chain.

PDF Certificate Gen
Institutional-grade A4 PDF with logo, signature, embedded QR, and monospace hash footer.

Bulk Issuance (CSV)
Batch issue certificates from CSV upload with progress monitoring and success/failure reporting.

Institutional Branding
Custom logo, signature upload, org name and domain — per-institution certificate identity.

DigiLocker Ready
PULL API endpoint with DigiLocker URI format — register on apisetu.gov.in, no code change needed.

Certificate Revocation
On-chain revocation with block timestamp. Only original issuer can revoke. Permanent record.

Role-Based Access
Supabase Auth + RLS. Issuers own their data; public can verify but not modify.

7 SECURITY ARCHITECTURE

- SHA-256 Hash Fingerprinting
- ECDSA Blockchain Signing
- Row-Level Security (RLS)
- Zod Schema Validation
- Server-Side Key Management
- HMAC-SHA256 DigiLocker Auth

Private keys are never exposed to the client. Parameterized queries prevent SQL injection. React's JSX escaping prevents XSS. Server Actions prevent CSRF.

8 RESULTS & PERFORMANCE

Issuance takes 5–15s on Amoy testnet (single tx per certificate). Mainnet (Polygon PoS) estimated **~0.001 POL ≈ <$0.001** per certificate. Supabase PostgreSQL with indexed `certificate_hash` and `issuer_id` supports horizontal scaling. Vercel deployment enables serverless auto-scaling for unlimited concurrent verifications. Verification latency < 2 seconds, 100% tamper detection accuracy, zero gas cost for verifiers, and ∞ blockchain lifespan.

9 FUTURE WORK

- **Polygon Mainnet:** Migrate from testnet to Polygon PoS mainnet for production credential issuance.
- **Merkle Tree Batch Issuance:** Bundle multiple hashes in a single transaction to cut gas costs for bulk scenarios.
- **Multi-Sig Wallets per Institution:** Each issuer gets independent multi-signature wallet for enhanced security.
- **Production DigiLocker Integration:** Register on apisetu.gov.in — API structure is complete, only config needed.
- **Real-Time Fraud Alerts:** Notify issuers when a certificate is verified, enabling fraud pattern detection.

10 CONCLUSION

Certitrust demonstrates that **certificate authenticity can be mathematically guaranteed** — not institutionally assumed. By anchoring SHA-256 hashes on Polygon's immutable blockchain, it decouples verification from any single server, company, or domain. Even if the website is cloned or shut down, every certificate remains verifiable on Polygonscan forever. With forensic OCR, bulk issuance, institutional branding, and a DigiLocker-ready API, Certitrust is **production-grade infrastructure** for the digital credential ecosystem.

CERTIFICATE LIFECYCLE

Issued (PDF) → Hashed (SHA-256) → Anchored (Polygon) → Verified (Public) → Revoke (Optional)

KEY PERFORMANCE METRICS

< 2s	100%	₹0	∞
Verification Latency	Tamper Detection	Verification Cost	Blockchain Lifespan
Zero gas	SHA-256 + OCR	Read-only blkchain	Immutable

DigiLocker Integration Architecture Complete
Awaiting apisetu.gov.in registration — zero code changes required for go-live

Live Demo
`certiproof-eta.vercel.app`
Scan to verify a real certificate or issue a demo credential. No account required for verification.

Built with Next.js · Solidity · Polygon · Supabase · Azure · Vercel